AENL 228: Measurement and Instrumentation for Energy Systems
Assignment I

Instructions:

1. Submit your work as a single, clearly scanned PDF of handwritten solutions.
2. Start each question on a fresh page or section with the problem number prominently labeled.
3. Explicitly state all standard assumptions at the beginning of each problem (e.g., ideal gas behavior, constant specific heats, negligible kinetic/potential energy changes, or steady-flow adiabatic conditions).
4. Show all intermediate derivations and formula substitutions. State thermodynamic property values alongside their source (e.g., steam tables, ideal gas relations) before numeric evaluation.
5. Retain at least three significant figures during calculations and report final answers with appropriate SI units (kJ, kPa, kW, kJ/kg-K).
6. Prior to submission, name the final PDF file in the exact designated structure: 'Entry Number_Full Name.pdf' (e.g., 2026ESZ8001_Rajeev_Awasthi.pdf)
7. Submission Deadline: 05:00 PM, 5th October, 2026

Question 1: A set of ten voltage measurements taken during a calibration test yields the values: 10.2 V, 10.5 V, 10.1 V, 10.3 V, 10.4 V, 10.2 V, 10.6 V, 10.3 V, 10.5 V, and 10.2 V. Calculate the arithmetic mean, the population variance (σ^2), and the sample standard deviation (s).

Question 2: For a batch of eleven pressure transducer readings given by 120 kPa, 125 kPa, 122 kPa, 125 kPa, 130 kPa, 121 kPa, 125 kPa, 128 kPa, 124 kPa, 126 kPa, and 127 kPa, determine the median, the mode, and the arithmetic mean.

Question 3: The following frequency distribution shows the daily power consumption (in kilowatt-hours, kWh) recorded at an experimental test rig over 50 days. Calculate the arithmetic mean power consumption.

Class Interval (kWh)	Frequency (f_i)
100 – 120	4
120 – 140	10
140 – 160	18
160 – 180	12
180– 200	6

Question 4: During an aerodynamic study, the velocity distribution of air passing through a wind tunnel section was measured. The results are grouped into the class intervals shown below. Determine the median velocity in meters per second.

Class Interval (m/s)	Frequency (f_i)
10 – 20	5
20 – 30	12
30 – 40	25
40 – 50	18
50 – 60	10

Question 5: The noise levels (in decibels, dB) generated by an array of cooling fans were monitored, resulting in the frequency table below. Find the modal noise level of the distribution.

Class Interval (dB)	Frequency (f_i)
50 – 60	3
60 – 70	8
70 – 80	22
80 – 90	15
90 – 100	7

Question 6: The time constants (in milliseconds) of 40 thermocouples subjected to a step temperature change are recorded in the grouped frequency table below. Calculate the variance (σ^2) and the standard deviation (σ) of the sample.

Class Interval (ms)	Frequency (f_i)
2.0 – 4.0	4
4.0 – 6.0	9
6.0 – 8.0	15
8.0 – 10.0	8
10.0– 12.0	4

Question 7: The pressure drops (in kilopascals, kPa) across a set of experimental flow-obstruction meters are measured and categorized into the class intervals given below. Calculate the arithmetic mean, variance, and standard deviation for this grouped data distribution.

Class Interval (kPa)	Frequency (f_i)
0 – 10	6
10 – 20	14
20 – 30	20
30 – 40	7
40 – 50	3

Question 8: The electric power dissipated in a resistor is calculated using $P = I^2R$. If the current is measured as $I = 5.0$ A ± 1% and the resistance is $R = 20\ \Omega \pm 0.5\%$, calculate the nominal power and the percentage uncertainty in the power determination using root-sum-square propagation.

Question 9: Error Propagation (Additive Function): The total length L of three sections in series is given by $L = L_1 + L_2 + L_3$. If $L_1 = 150.0 \pm 0.05$ cm, $L_2 = 200.0 \pm 0.08$ cm, and $L_3 = 120.0 \pm 0.04$ cm, calculate the total length and its absolute uncertainty.

Question 10: The mass flow rate through an orifice meter is given by $\dot{m} = C\ A \sqrt{(2\rho\Delta p)}$. Given $C = 0.92 \pm 0.5\%$, $A = 0.002\ m^2 \pm 0.1\%$, density $\rho = 1.2\ kg/m^3 \pm 1.0\%$, and pressure drop $\Delta p = 1200$ Pa ± 0.8%, determine the percentage uncertainty in the mass flow rate.

Question 11: A vibration displacement sensor is tested 16 times, yielding a sample standard deviation of 0.04 mm. Calculate the standard deviation of the mean (σ_m) and establish the 95% confidence interval for the true mean displacement using the normal distribution ($z = 1.96$).

Question 12: How many independent measurements must be taken to ensure that the sample mean estimates the population mean within ± 0.5 mm with a 95% confidence level, given that preliminary tests show a standard deviation of 2.0 mm?

Question 13: A U-tube manometer using mercury ($\rho_m = 13600\ kg/m^3$) is connected to a gas pipeline. If the deflection between the two columns is $h = 250$ mm and atmospheric pressure is 101.3 kPa, calculate the gauge pressure and absolute pressure in the pipeline ($g = 9.81\ m/s^2$).

Question 14: Strain Measurement (Resistance Strain Gage): A foil strain gage with a gage factor $GF = 2.05$ is mounted on a steel cantilever beam. When a load is applied, the resistance changes by $\Delta R = 0.0164\ \Omega$ from a baseline resistance $R = 120\ \Omega$. Calculate the strain (ε) in microstrain ($\mu\varepsilon$).

Question 15: A Venturi meter with an inlet diameter $d_1 = 10$ cm and throat diameter $d_2 = 5$ cm is used to measure water flow ($\rho = 1000\ kg/m^3$). If the differential pressure across the meter is 50 kPa and the discharge coefficient is $C_d = 0.98$, calculate the volumetric flow rate in m^3/s.

Question 16: A platinum resistance thermometer has a resistance of 100.0 Ω at 0°C and a temperature coefficient of resistance $\alpha = 0.00385\ (°C)^{-1}$. If the measured resistance in an oven is 138.5 Ω, calculate the measured temperature.

Question 17: A sling psychrometer reads a dry-bulb temperature of 30°C and a wet-bulb temperature of 22°C. Given saturation vapor pressures $P_{g@30°C}$ = 4.246 kPa and $P_{g@22°C}$ = 2.645 kPa, and psychrometric constant $\gamma = 0.000666\ K^{-1}$ at P = 101.3 kPa, calculate the partial pressure of water vapor and the relative humidity.

Question 18: A capillary tube of radius r = 0.5 mm and length L = 0.3 m is used to measure the dynamic viscosity of an oil. If a volumetric flow rate $Q = 1.2 \times 10^{-6}\ m^3/s$ is driven by a pressure drop Δp = 15 kPa, calculate the dynamic viscosity (μ) using Hagen-Poiseuille flow ($\mu = (\pi\ \Delta p\ r^4) / (8\ L\ Q)$).

Question 19: In an internal combustion engine exhaust analysis, the dry volumetric concentration of carbon dioxide is measured as 12.5% and carbon monoxide as 1.2%. Calculate the total carbon-bearing exhaust gas volume percentage and the stoichiometric excess air ratio if the ideal dry volume sum is 100%.

Question 20: Temperature Measurement Error (Conduction Error): A mercury-in-glass thermometer is inserted into an air duct to measure gas temperature. The duct wall is at 20°C, while the true gas temperature is 100°C. If the thermal properties result in a steady indicated temperature of 92°C, calculate the absolute temperature measurement error and percentage error.

Question 21: A glass electrode pH meter immersed in an aqueous wastewater solution at 25°C generates a measured potential of 0.209 V relative to a standard reference electrode. Given the Nernst slope factor of 0.05916 V per pH unit at 25°C ($E = E_0 - 0.05916 \cdot pH$), calculate the pH of the solution.

Question 22: A solid hydrometer of mass m = 25 g and stem cross-sectional area A = 0.5 cm^2 floats in an unknown liquid. When immersed to a reference mark, the displaced volume is V_0 = 24.5 cm^3. Calculate the density of the test liquid in g/cm^3.

Question 23: A thermocouple behaving as a first-order system has a time constant τ = 4.5 s. Initially at 25°C, it is suddenly exposed to a fluid bath at 150°C. Calculate the temperature indicated by the thermocouple after 5 s and the time required to reach 90% of the steady-state value.

Question 24: A temperature sensor with a time constant $\tau = 2.0$ s is subjected to a harmonic temperature fluctuation of frequency $f = 0.05$ Hz. Calculate the phase lag angle in degrees and the time delay in seconds.

Question 25: A pressure transducer modeled as a second-order spring-mass-damper system has a natural frequency $f_n = 2000$ Hz and a damping ratio $\zeta = c/c_c = 0.6$. Calculate the damped natural frequency (f_r) of the ringing oscillation.

Question 26: A second-order accelerometer with natural frequency $\omega_n = 500$ rad/s and damping ratio $\zeta = 0.707$ is subjected to a sinusoidal vibration input at an excitation frequency $\omega = 350$ rad/s. Calculate the amplitude ratio (magnification factor).

Question 27: A voltage source has an open-circuit internal voltage $E = 24$ V and an internal resistance $R_i = 500\ \Omega$. A voltmeter with input resistance $R_m = 4500\ \Omega$ is connected across the terminals. Calculate the voltage indicated by the meter and the percentage loading error compared to the true open-circuit voltage.

Question 28: Classify the three main types of measurement errors (gross, systematic, and random). For each error type, identify one primary source of occurrence and state how its effect can be minimized or corrected.

Question 29: Discuss the concept of instrument sensitivity and resolution. Does high resolution guarantee high measurement accuracy? Justify your answer using the relationship between sensor sensitivity and electronics capability.

Question 30: Define the terms accuracy, precision, and stability. Using the target/bullseye analogy, explain how an instrument can possess high precision yet low accuracy.